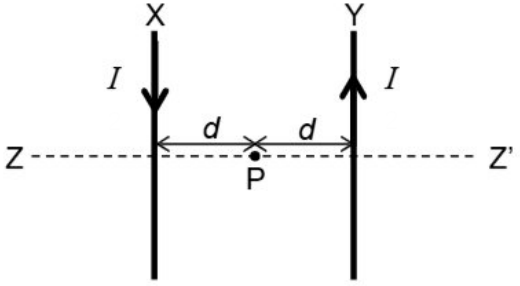
25 Two infinitely long straight conductors X and Y are placed parallel to each other. Each conductor carries a current I in opposite directions to each other. Point P is equidistant from either conductor. The magnetic flux density at P due to each conductor is B.

X is fixed while Y is rotated about the axis ZZ' until it is perpendicular to X and its current is into the plane of the paper.



What is the final resultant magnetic flux density at P in terms of B?

A 0 **B** B **C** $\sqrt{2}B$ **D** $2B$